

Separable Equations



Order and verifying solutions

- 1 State the order of the differential equation $y'' + 3y' - 2y = \sin x$.
 - 2 Verify that $y = e^{2x}$ is a solution of $y' - 2y = 0$.
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Recognizing and solving separable equations

- 3 Solve $\frac{dy}{dx} = xy$.
 - 4 Solve $\frac{dy}{dx} = \frac{y^2}{x}$ for $x > 0$.
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Initial value problems

- 5 Solve $\frac{dy}{dx} = 3x^2y$, $y(0) = 2$.
 - 6 Solve $\frac{dy}{dx} = -2xy$, $y(0) = 5$.
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Application: radioactive decay and population growth

- 7 A radioactive substance decays according to $\frac{dA}{dt} = -kA$. If $A(0) = 100$ g and $A(5) = 50$ g, find k .
 - 8 A population grows according to $\frac{dP}{dt} = 0.03P$, with $P(0) = 200$. Find $P(10)$.
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Further practice: separable equations

- 9 Solve $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$.
- 10 Solve $y' = y \cos x$, $y(0) = 1$.